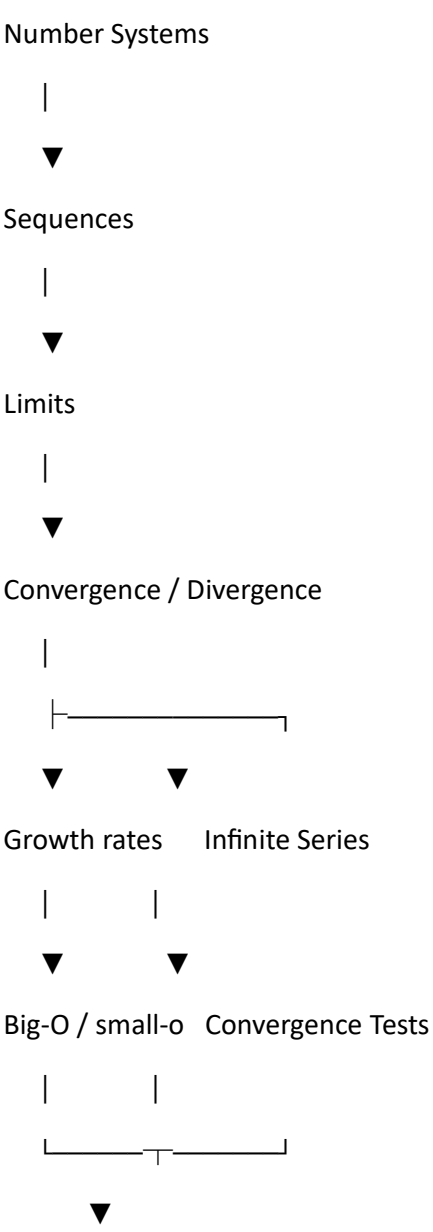
Interview Notes — Numbers, Sequences & Series

These notes are based on your uploaded **Chapter 10: "Numbers, Sequences, and Series"**. I've applied an **80/20 interview filter**: focus heavily on concepts that help you reason about limits, convergence, algorithmic growth, numerical computation, probability, and ML optimization; keep deeper theorem machinery as secondary. The chapter itself emphasizes that sequences and their limits provide foundations for quantities used throughout ML, including gradient descent, differentiation, integration, probability, and statistics.

---

1. The Big Picture

Think of the chapter as one chain:



ML / Numerical Analysis

|

├— Gradient Descent

├— Numerical Approximation

├— Probability

├— Integration

└— Algorithm Complexity

The most important interview idea is:

A sequence describes values as the index grows. A limit describes long-term behavior. A series is the accumulation of sequence terms.

---

2. Number Systems — 80/20

The chapter introduces five important number classes:

$$\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$$

with

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$$

For this chapter, the first four are the focus.

2.1 Natural Numbers

Definition

$$\mathbb{N} = \{1, 2, 3, \dots\}$$

The source uses the convention excluding 0, while noting that some definitions include 0. It separately denotes

$$\mathbb{N}_0 = \{0, 1, 2, \dots\}$$

when 0 is included.

Example

1, 2, 3, 4, 5, ...

Interview relevance

Low individually.

Know it because sequences are normally indexed using natural numbers:

$$a_1, a_2, a_3, \ldots$$

---

3. Integers

Definition

$$\mathbb{Z} = \{\ldots, -, 2, -, 1, 0, 12 \ldots\}$$

Integers contain additive inverses:

$$n \in \mathbb{Z} \Rightarrow -n \in \mathbb{Z}$$

and are closed under addition:

$$a, b \in \mathbb{Z} \Rightarrow a + b \in \mathbb{Z}$$

The source specifically highlights these two properties.

Diagram

<---- negative ----|---- positive ---->

... -3 -2 -1 0 1 2 3 ...

Example

$$-7 + 12 = 5$$

Both input and output are integers.

---

4. Rational Numbers

Definition

A rational number can be written as

$$\mathbb{Q} = \left\{\frac{p}{q}, :, pq \in \mathbb{Z}\, q \neq 0\right\}$$

Examples:

$$\frac{1}{2}, \frac{7}{3}, -\frac{5}{4}, 5$$

The chapter notes that rational numbers are closed under multiplication and every nonzero rational has a multiplicative inverse.

Decimal representation

A rational number has either:

Finite decimal

OR

Repeating decimal

Examples:

$$\frac{1}{2} = 0.5$$
$$\frac{1}{3} = 0.3333\ldots$$

The source also points out that decimal representations aren't always unique:

$$1.0 = 0.999\ldots$$

---

5. Real Numbers

Definition / intuition

Real numbers can be visualized as every point on the real number line.

<--->

-∞ -2 -1 0 1 2 +∞

The chapter describes real numbers as including rational and irrational numbers and notes that unlike $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$, $\mathbb{R}$is **uncountable**.

Rational vs irrational

Real Numbers $\mathbb{R}$

/ \

Rational Irrational

ℚ |

1/2, 0.75 π, √2, e

Interview takeaway

Know:

- rational → finite/repeating decimal
- irrational → infinite non-repeating decimal
- real numbers include both
- real numbers are uncountable

---

6. Countable vs Uncountable

This is **secondary interview knowledge**, but understand the intuition.

Countable

A set is countable if its elements can be associated with natural-number indices.

Examples from the chapter:

$$\mathbb{N}, \mathbb{Z}, \mathbb{Q}$$

Uncountable

$$\mathbb{R}$$

The chapter demonstrates this using **Cantor's diagonal argument**.

Intuition

Suppose someone claims:

All real numbers in [0,1)

a1 = 0.a11 a12 a13 ...

a2 = 0.a21 a22 a23 ...

a3 = 0.a31 a32 a33 ...

...

Take the diagonal:

a11

a22

a33

a44

...

Change every diagonal digit.

You construct a new number:

b = 0.b1 b2 b3 b4 ...

such that:

$$b_i \neq a_{ii}$$

Therefore $b$differs from:

a1 in digit 1

a2 in digit 2

a3 in digit 3

...

So it cannot be anywhere in the supposed enumeration.

Interview question

Q: Why can't all real numbers be represented exactly by computer floating-point numbers?

Because:

Real numbers → uncountably infinite

Floating-point representations → finite

So floating-point arithmetic necessarily represents only approximations.

The chapter explicitly makes this observation in the context of π, e, and numerical approximation.

---

7. Sequences ⭐ ⭐ ⭐

This is one of the **highest-priority topics**.

Definition

A sequence is an enumeration of mathematical objects.

For this chapter:

$$\{a_n\}_{n=1}^{\infty}$$

where

$$a_n \in \mathbb{R}$$

The chapter emphasizes that sequences describe long-term behavior and connects them directly to ML optimization.

Example

$$a_n = \frac{1}{n}$$

produces

n: 1 2 3 4 5 ...

a_n: 1 0.5 0.333 0.25 0.2 ...

---

8. Subsequence

Definition

Given

$$a_1, a_2, a_3, a_4, \ldots$$

choose indices

$$n_1 < n_2 < n_3 < \cdots$$

Then

$$a_{n_1}, a_{n_2}, a_{n_3}, \ldots$$

is a subsequence.

The chapter describes this intuitively as **throwing elements away from a sequence**.

Example

Original:

1 2 3 4 5 6 7 8 9 ...

Subsequence:

2 4 6 8 10 ...

because

$$n_k = 2k$$

Interview point

A subsequence **preserves order** but can skip elements.

---

9. Convergence ⭐⭐⭐

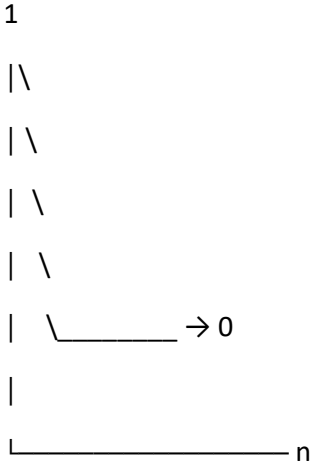
This is probably the most important mathematical concept in the chapter.

Intuition

A sequence converges to $a$if its values eventually get arbitrarily close to a.

Example:

$$a_n = \frac{1}{n}$$



Therefore:

$$\boxed{\lim_{n\to\infty}\frac{1}{n}=0}$$

---

10. Formal Definition of Convergence ⭐ ⭐ ⭐

The sequence $\{a_n\}$converges to $a$if:

$$\forall\epsilon>0,\ \exists n_0\in\mathbb{N}$$

such that

$$|\ a_n-a\ |<\epsilon$$

for every

$$n>n_0$$

The source gives this as the formal definition of convergence.

Interview-friendly interpretation

You choose how close you want me to get

↓

ε

↓

I give you some sufficiently large n_o

↓

After n_o, every value stays within ε of the limit

Diagram

ε

<---------->

[a]

|

————————┼———————— real line

a-ε | a+ε

Sequence eventually enters this interval

and never leaves it.

---

11. Example: Prove $1/n \to 0$

We need:

$$\left|\frac{1}{n} - 0\right| < \epsilon$$

So:

$$\frac{1}{n} < \epsilon$$

which means

$$n > \frac{1}{\epsilon}$$

Choose

$$n_0 = \left\lfloor \frac{1}{\epsilon} \right\rfloor + 1$$

Then for all $n > n_0$,

$$\frac{1}{n} < \epsilon$$

Therefore:

$$\boxed{\lim_{n\to\infty} \frac{1}{n} = 0}$$

This exact example appears in the source.

---

12. Why Convergence Matters for ML ⭐ ⭐ ⭐

This is an **extremely important interview connection**.

The chapter connects convergence with:

- differentiation
- integration
- expected values
- probability
- stochastic gradient descent
- gradient descent itself

Gradient descent connection

Suppose:

$$w_{t+1} = w_t - \eta \nabla L(w_t)$$

Then:

w_0

↓

w_1

↓

w_2

↓

w_3

↓

...

↓

hopefully approaches an optimum

So:

$$\{w_t\}$$

is a sequence.

The question becomes:

Does the sequence of model parameters converge?

And possibly:

$$L(w_t) \to L^*$$

Interview answer

Gradient descent can be viewed as generating a recursively defined sequence of model parameters. Convergence asks whether those iterates approach a limiting point or objective value.

The chapter explicitly describes gradient descent as a recursively defined sequence of model weights.

---

13. Properties of Convergent Sequences ⭐ ⭐

Suppose:

$$a_n \to a$$

and

$$b_n \to b$$

Then:

Addition

$$a_n + b_n \to a + b$$

Scalar multiplication

$$ca_n \to ca$$

Multiplication

$$a_n b_n \to ab$$

Reciprocal

If $a \neq 0$and $a_n \neq 0$,

$$\frac{1}{a_n} \to \frac{1}{a}$$

These properties are explicitly given in the chapter.

Example

If

$$a_n = \frac{1}{n}$$

and

$$b_n = 2 + \frac{1}{n}$$

then

$$a_n \to 0$$

and

$$b_n \to 2$$

Therefore:

$$a_n + b_n \to 2$$

and

$$a_n b_n \to 0$$

---

14. Continuity and Limits

An important rule:

$$\boxed{\lim_{n\to\infty} f(x_n) = f\left(\lim_{n\to\infty} x_n\right)}$$

when f is continuous at the limit.

The chapter describes continuity in terms of interchangeability of limits and functions.

Example

If

$$x_n \to 2$$

then for continuous $f(x) = x^2$:

$$x_n^2 \to 4$$

because:

$$\lim x_n^2 = (\lim x_n)^2 = 2^2 = 4$$

---

15. Transfer Principle ⭐ ⭐

If

$$a_n \geq \alpha$$

for all n, and

$$a_n \to a$$

then

$$a \geq \alpha$$

The source calls this the **transfer principle**.

Important trap

If

$$a_n > \alpha$$

for every n, it does **not** necessarily mean

$$\lim a_n > \alpha$$

Example:

$$a_n = \frac{1}{n} > 0$$

but

$$\lim a_n = 0$$

So:

Every term > 0

≠

Limit > 0

---

16. Squeeze Principle ⭐ ⭐ ⭐

Suppose:

$$a_n \leq b_n \leq c_n$$

and

$$a_n \to L,\ c_n \to L$$

then:

$$\boxed{b_n \to L}$$

The chapter gives this as the squeeze principle.

Diagram

c_n ——————————→ L

\ /

\ b_n /

\ /

a_n ——————————→ L

Example

Suppose:

$$-\frac{1}{n} \leq a_n \leq \frac{1}{n}$$

Since:

$$-\frac{1}{n} \to 0$$

and

$$\frac{1}{n} \to 0$$

we conclude:

$$a_n \to 0$$

Interview memory

Trap the sequence between two sequences with the same limit.

---

17. Famous Limits — Memorize the Useful Ones ⭐⭐⭐

The source gives several important growth/convergence examples.

Exponential

For $x \geq 0$:

$$\lim_{n\to\infty} x^n = \begin{cases} 0 & 0 \leq x < 1 \\ 1 & x = 1 \\ \infty & x > 1 \end{cases}$$

Example

$$\left(\frac{1}{2}\right)^n \to 0$$

but

$$2^n \to \infty$$

---

nth Root

For $x \geq 0$:

$$\lim_{n\to\infty} x^{1/n} = \begin{cases} 1 & x > 0 \\ 0 & x = 0 \end{cases}$$

Example:

$$1000000^{1/n} \to 1$$

---

18. Growth Rates ⭐ ⭐ ⭐

This is **very important for software/ML interviews**.

The chapter compares:

$$\log n\,,\ n,\ 2^n,\ n!$$

and gives:

$$\frac{\log n}{n} \to 0$$
$$\frac{n}{2^n} \to 0$$
$$\frac{2^n}{n!} \to 0$$

So the growth hierarchy is:

log n

↓

n

↓

2^n

↓

n!

meaning each later function eventually grows faster.

Interview example

Compare:

```
for (int i = 1; i <= n; i++) {
   // O(1)
```

}

→

$$O(n)$$

versus nested loops:

```
for (int i = 0; i < n; i++) {
   for (int j = 0; j < n; j++) {
      // O(1)
   }
}
```

→

$$O(n^2)$$

As $n$becomes large, n^2grows much faster than n.

---

19. Surprising Limit: $n^{1/n}$

The source gives:

$$\boxed{\lim_{n\to\infty} n^{1/n} = 1}$$

The logarithm trick explains it:

$$n^{1/n} = e^{\ln(n)/n}$$

Since:

$$\frac{\ln n}{n} \to 0$$

we get:

$$e^0 = 1$$

Interview trick

Whenever you see:

$$x_n^{y_n}$$

and both exponent/base behavior looks difficult, consider taking logarithms.

---

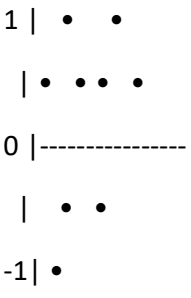
20. Divergent Sequences ⭐ ⭐ ⭐

A sequence that does not converge is divergent.

Example:

$$a_n = \sin(n)$$

It keeps oscillating rather than approaching one number. The source uses this example.



Therefore:

$$\sin(n) \not\to L$$

for a single real L.

---

21. Divergence to Infinity

A sequence is ∞-divergent if eventually it exceeds every arbitrarily large number.

Examples:

$$a_n = n$$
$$a_n = n \log n$$

Then:

$$a_n \to \infty$$

The chapter explicitly connects this behavior to algorithm runtime.

---

22. Big-O ⭐ ⭐ ⭐ ⭐ ⭐

This is probably the **highest interview-value concept outside convergence itself**.

Definition

The chapter defines:

$$b_n = O(a_n)$$

if there exists a constant $C > 0$such that:

$$| b_n | \leq C | a_n |$$

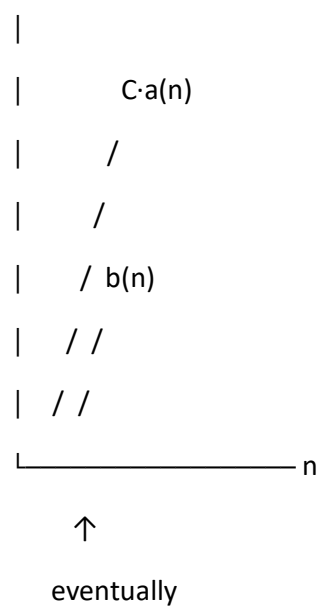
for sufficiently large n.

Intuition

Big-O means:

Eventually, $b_n$does not grow faster than a constant multiple of a_n.

Diagram



---

23. Big-O Example

Suppose:

$$f(n) = 3n + 10$$

We claim:

$$f(n) = O(n)$$

Because for sufficiently large n:

$$3n + 10 \leq 4n$$

whenever

$$n \geq 10$$

Therefore:

$$\boxed{3n + 10 = O(n)}$$

Important interview point

Big-O ignores:

- constant factors
- lower-order terms

So:

$$3n + 10$$

and

$$100n + 500$$

are both:

$$O(n)$$

---

24. Small-o ⭐ ⭐ ⭐

The chapter defines:

$$b_n = o(a_n)$$

if for every

$$\epsilon > 0$$

there exists Nsuch that:

$$| b_n | \leq \epsilon | a_n |$$

for every $n > N$.

Key intuition

$$\boxed{b_n = o(a_n)}$$

means:

$b_n$becomes negligible compared with a_n.

Equivalent intuition:

$$\frac{b_n}{a_n} \to 0$$

when the ratio is well-defined.

Example

$$\log n = o(n)$$

because:

$$\frac{\log n}{n} \to 0$$

Big-O vs little-o

O:

b grows no faster than a constant multiple of a

o:

b becomes insignificant compared with a

Critical distinction

$$o(n) \Rightarrow O(n)$$

but generally:

$$O(n) \Rightarrow\!/\, o(n)$$

For example:

$$n = O(n)$$

but

$$n \neq o(n)$$

because

$$\frac{n}{n} = 1 \rightarrow\!/\, 0$$

---

25. Why Big-O Matters in Your Interviews

For your Java/backend + AI/ML direction, think:

Backend

Database query

↓

Query complexity

↓

Index / no index

↓

Latency as data grows

ML

Training data n

↓

Algorithm

↓

Training complexity

↓

Inference complexity

↓

Memory growth

Example:

A linear scan:

$$O(n)$$

Hash lookup:

$$O(1)$$

Balanced tree lookup:

$$O(\log n)$$

Naive pairwise comparison:

$$O(n^2)$$

This is exactly why asymptotic growth matters.

---

26. Why $\mathbb{Q}$ Is Not Enough

A subtle but important idea from the chapter:

Consider

$$a_n = \left(1 + \frac{1}{n}\right)^n$$

Every a_n is rational.

But:

$$\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n = e$$

and $e$is irrational.

Therefore rational numbers aren't closed under taking limits.

The chapter uses this to motivate the idea that real numbers can be obtained by **closing rational numbers under limits**.

Mental model

$\mathbb{N}$

| add negatives + 0

▼

$\mathbb{Z}$

| add multiplicative inverses

▼

$\mathbb{Q}$

| close under limits

▼

$\mathbb{R}$

This is much more important conceptually than memorizing the construction.

---

27. Numerical Approximation of e

The chapter gives two ways to approximate e.

Method 1

$$e = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n$$

Method 2

$$e = \sum_{n=0}^{\infty} \frac{1}{n!}$$

The second representation converges rapidly because $n!$grows very quickly. The source also points out numerical stability/convergence considerations when computing the first expression.

Implementation

```
import math


def approximate_e(n):
    return (1 + 1/n) ** n


def approximate_e_series(n):
    total = 0.0

    for k in range(n):
        total += 1 / math.factorial(k)

    return total


print(approximate_e(100000))
print(approximate_e_series(15))
```

Interview lesson

Don't just ask:

"Does it converge?"

Also ask:

How fast does it converge, and is the numerical computation stable?

That distinction is extremely useful in ML.

---

28. Infinite Series ⭐⭐⭐⭐⭐

A series is:

$$\sum_{n=1}^{\infty} a_n$$

The important idea:

A series is not "adding infinity terms directly." It is defined through the limit of its partial sums.

The source defines:

$$S_N = \sum_{n=1}^{N} a_n$$

and

$$\boxed{\sum_{n=1}^{\infty} a_n = \lim_{N\to\infty} S_N}$$

if the limit exists.

Diagram

$a_1 + a_2 + a_3 + a_4 + \ldots$

↓

$S_1 = a_1$

$S_2 = a_1+a_2$

$S_3 = a_1+a_2+a_3$

$S_4 = a_1+a_2+a_3+a_4$

...

↓

lim S_n

This distinction is **very important for interviews**.

---

29. Series vs Sequence

Common interview trap.

Sequence

$$a_1, a_2, a_3, \ldots$$

is a list of values.

Series

$$a_1 + a_2 + a_3 + \cdots$$

is the sum of those values.

Example

Sequence:

$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots$$

Series:

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots$$

The sequence converges:

$$\frac{1}{n} \to 0$$

but the series diverges:

$$\sum_{n=1}^{\infty} \frac{1}{n} = \infty$$

This is one of the most important traps in the chapter.

---

30. Necessary Condition for Series Convergence ⭐⭐⭐⭐⭐

If

$$\sum_{n=1}^{\infty} a_n$$

converges, then necessarily:

$$\boxed{a_n \to 0}$$

The source explicitly states this proposition.

But the reverse is FALSE.

$$a_n \to 0$$

does **not** imply:

$$\sum a_n$$

converges.

Counterexample

$$a_n = \frac{1}{n}$$

Then:

$$a_n \to 0$$

but:

$$\sum_{n=1}^{\infty} \frac{1}{n} = \infty$$

Interview answer

For a series to converge, its terms must approach zero, but that condition is not sufficient.

Memorize this.

---

31. Geometric Series ⭐ ⭐ ⭐ ⭐ ⭐

One of the most important series.

$$\sum_{n=0}^{\infty} q^n$$

If

$$| q | < 1$$

then:

$$\boxed{\sum_{n=0}^{\infty} q^n = \frac{1}{1-q}}$$

The source gives the convergence condition and formula.

Example

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots$$

Here:

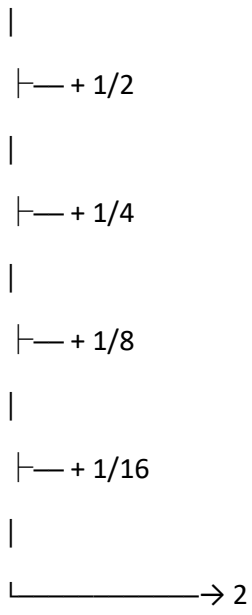
$$q = \frac{1}{2}$$

so:

$$S = \frac{1}{1 - \frac{1}{2}} = 2$$

Diagram

1



Interview rule

|q| < 1 → converges

|q| ≥ 1 → diverges

---

32. Geometric Series Implementation

```
def geometric_sum(q, n):
    total = 0.0
    term = 1.0

    for _ in range(n):
        total += term
        term *= q

    return total

print(geometric_sum(0.5, 20))
```

Expected value approaches:

Production/ML lesson

This is a simple example of iterative approximation:

exact infinite quantity

↓

finite computation

↓

partial sum

↓

approximation

---

33. Harmonic Series ⭐ ⭐ ⭐ ⭐ ⭐

$$\boxed{\sum_{n=1}^{\infty} \frac{1}{n} = \infty}$$

It diverges.

The source proves this using grouping of terms and notes that partial sums grow approximately like:

$$\log N + \gamma$$

Important surprise

1/n → 0

but

1 + 1/2 + 1/3 + 1/4 + ... → ∞

This is the classic interview example showing:

$$a_n \to 0$$

is not sufficient for series convergence.

---

34. Alternating Harmonic Series ⭐ ⭐ ⭐

$$\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{n}$$

converges to:

$$\boxed{\ln 2}$$

The source identifies it as conditionally convergent.

It looks like:

1 - 1/2 + 1/3 - 1/4 + 1/5 - ...

The positive and negative terms partially cancel each other.

---

35. Absolute vs Conditional Convergence ⭐ ⭐ ⭐ ⭐ ⭐

Absolute convergence

If

$$\sum |a_n|$$

converges, then

$$\sum a_n$$

is **absolutely convergent**.

Conditional convergence

If

$$\sum a_n$$

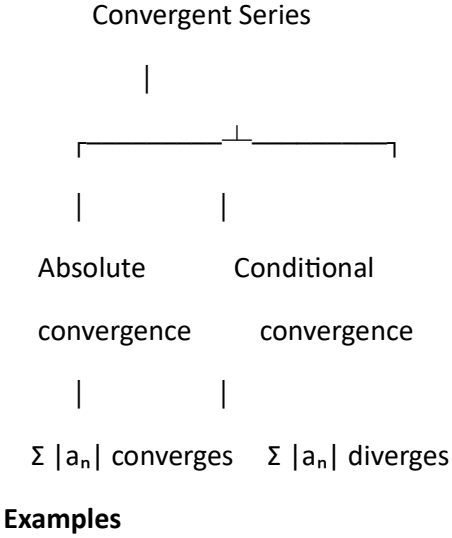
converges but

$$\sum |a_n|$$

doesn't, it is **conditionally convergent**.

These definitions are explicitly given in the source.

Diagram



Examples

Geometric:

$$\sum \left(-\frac{1}{2}\right)^n$$

is absolutely convergent.

Alternating harmonic:

$$\sum (-1)^{n+1} \frac{1}{n}$$

is conditionally convergent.

---

36. Why Absolute Convergence Matters

If a series is absolutely convergent, rearranging its terms does not change its sum.

The source gives:

$$\sum a_{\sigma(n)} = \sum a_n$$

for arbitrary permutations σ, when the original series is absolutely convergent.

Mental model

Absolute convergence

↓

safe rearrangement

↓

same sum

Conditional convergence is different.

The chapter discusses the Riemann rearrangement theorem: a conditionally convergent series can be rearranged to produce different sums.

Interview takeaway

Absolute convergence gives much stronger stability under rearrangement than conditional convergence.

---

37. Permutation

A permutation is a bijective mapping from a set to itself.

Informally:

1 2 3 4 5 6

↓ ↓ ↓ ↓ ↓ ↓

2 1 4 3 6 5

The source uses this concept to formalize rearrangement of series.

Interview relevance

Low individually.

Know it because it explains:

Why does changing the order of terms sometimes change an infinite sum?

---

38. Direct Comparison Test ⭐ ⭐ ⭐ ⭐

The idea:

Compare an unknown series with a known series.

If:

$$0 \leq a_n \leq b_n$$

and

$$\sum b_n$$

converges, then:

$$\sum a_n$$

converges.

Conversely, if:

$$0 \leq b_n \leq a_n$$

and

$$\sum b_n$$

diverges, then:

$$\sum a_n$$

diverges.

This is the direct comparison principle described in the chapter.

Example

Consider:

$$\sum_{n=1}^{\infty} \frac{1}{n^2+1}$$

Since:

$$0 < \frac{1}{n^2+1} < \frac{1}{n^2}$$

and

$$\sum \frac{1}{n^2}$$

converges, therefore:

$$\boxed{\sum \frac{1}{n^2+1} \text{ converges}}$$

---

39. p-Series Pattern ⭐⭐⭐⭐⭐

A very useful interview pattern is:

$$\sum_{n=1}^{\infty} \frac{1}{n^{\alpha}}$$

The source uses comparison with the harmonic series and inverse-square series to establish the key threshold.

The important result to remember:

$$\boxed{\sum \frac{1}{n^{\alpha}} \begin{cases} \text{converges} & \alpha > 1 \\ \text{diverges} & \alpha \leq 1 \end{cases}}$$

Examples:

$$\sum \frac{1}{n}$$

diverges.

$$\sum \frac{1}{n^2}$$

converges.

$$\sum \frac{1}{n^3}$$

converges.

---

40. Root Test ⭐ ⭐ ⭐

For:

$$\sum a_n$$

look at:

$$\sqrt[n]{|\, a_n \,|}$$

If eventually:

$$\sqrt[n]{|\, a_n \,|} < 1$$

then the series is absolutely convergent.

The source specifically uses the root test to establish convergence of:

$$\sum \frac{1}{n!}$$

Interview intuition

Root test is particularly useful when the terms contain:

n-th powers

exponentials

factorials / exponential-like growth

---

41. Alternating Series Test ⭐ ⭐ ⭐ ⭐

Suppose:

$$\sum(-1)^n a_n$$

where $a_n \geq 0$.

If:

1. $a_n$decreases monotonically
2. $a_n \to 0$

then the alternating series converges.

The source calls this the **alternating test / Leibniz criterion**.

Example

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$$

Here:

$$a_n = \frac{1}{n}$$

and:

$$\frac{1}{n+1} < \frac{1}{n}$$

and:

$$\frac{1}{n} \to 0$$

Therefore it converges.

---

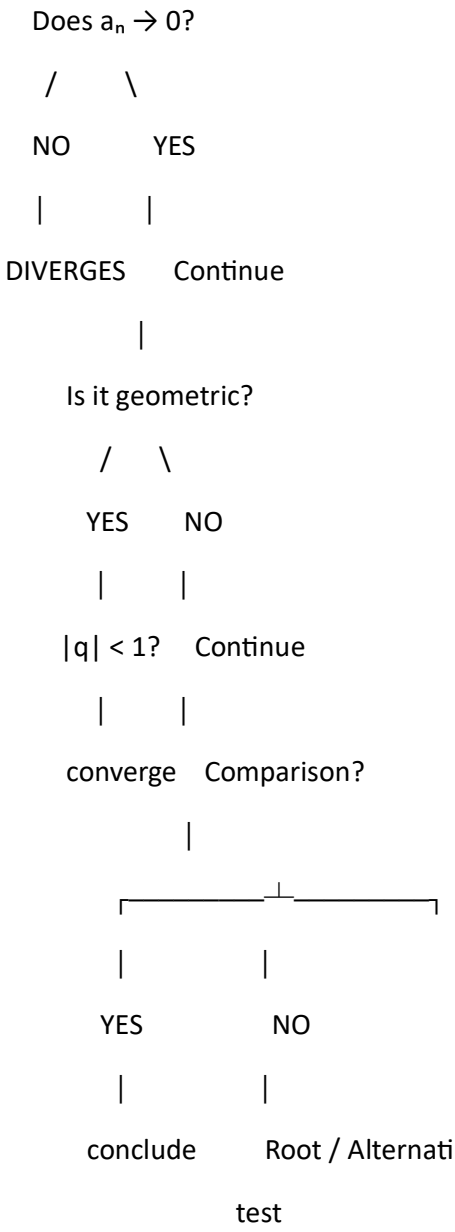
42. Series Convergence Decision Tree ⭐ ⭐ ⭐ ⭐ ⭐

For interviews, memorize this:

Given Σ a_n

|

▼



Critical rule

If:

$$a_n \not\to 0$$

then immediately:

$$\boxed{\sum a_n \text{ diverges}}$$

But:

$$a_n \to 0$$

doesn't finish the problem.

---

43. Basel Problem

The source gives the famous identity:

$$\boxed{\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}}$$

Interview relevance

Low.

Don't spend significant preparation time deriving it.

Know it as a famous example of a convergent series.

---

44. Euler's Number

The chapter gives two representations:

$$e = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n$$

and

$$e = \sum_{n=0}^{\infty} \frac{1}{n!}$$

ML connection

*e*appears everywhere:

exponential functions

↓

logarithms

↓

softmax

↓

log-loss

↓

probability models

↓

optimization

So understanding convergence of exponential-related sequences is useful groundwork.

---

45. Cauchy Product — Advanced ⭐⭐

If:

$$A = \sum_{k=0}^{\infty} a_k$$

and

$$B = \sum_{n=0}^{\infty} b_n$$

then their product can be represented using:

$$\boxed{\sum_{n=0}^{\infty} \left(\sum_{k=0}^{n} a_k \, b_{n-k} \right)}$$

This is called the **Cauchy product**. The chapter presents it through Mertens' theorem, under the condition that both series converge and at least one converges absolutely.

Diagram

Think of multiplication as a grid:

b0 b1 b2 b3

```
    ┌─────┬─────┬─────┬─────
a0  |a0b0 |a0b1 |a0b2 |a0b3
    ├─────┼─────┼─────┼─────
a1  |a1b0 |a1b1 |a1b2 |a1b3
    ├─────┼─────┼─────┼─────
a2  |a2b0 |a2b1 |a2b2 |a2b3
    ├─────┼─────┼─────┼─────
```

Then group diagonally:

diagonal 0 → a0b0

diagonal 1 → a0b1 + a1b0

diagonal 2 → a0b2 + a1b1 + a2b0

...

Hence:

$$c_n = \sum_{k=0}^{n} a_k \, b_{n-k}$$

Interview priority

Know the idea, not the proof.

---

46. Implementation: Partial Sums

For practical ML/numerical programming, infinite series are approximated with finite partial sums.

```
def partial_sum(term, N):
    total = 0.0
```

```
    for n in range(1, N + 1):
        total += term(n)

    return total


# Harmonic partial sum
result = partial_sum(lambda n: 1 / n, 1000)

print(result)
```

This computes:

$$S_{1000} = \sum_{n=1}^{1000} \frac{1}{n}$$

It doesn't calculate the infinite series directly.

---

47. Implementation: Detecting Convergence Numerically

A simple numerical approximation:

```
def approximate_series(term, tolerance=1e-8, max_terms=1_000_000):
    total = 0.0

    for n in range(1, max_terms + 1):
        value = term(n)
        total += value

        if abs(value) < tolerance:
            return total, n
```

return total, max_terms

Important warning

This is **not a mathematical proof of convergence**.

Why?

Because:

$$a_n \to 0$$

doesn't imply:

$$\sum a_n$$

converges.

For example:

term = lambda n: 1/n

The individual terms eventually become tiny, but the harmonic series still diverges.

---

48. Better Numerical Thinking

For numerical ML, ask three separate questions:

1. Does it converge?

↓

2. How fast does it converge?

↓

3. Is the computation numerically stable?

The chapter makes this distinction using approximation of e: one representation can have slow convergence or numerical instability, while the series representation can converge faster and be more numerically convenient.

This is an excellent **interview-level engineering mindset**.

---

49. ML Connections You Should Be Able to Explain

Connection 1 — Gradient Descent

Weights

↓

w_0

↓

w_1

↓

w_2

↓

...

This creates a sequence.

Question:

$$w_n \to w^*?$$

The chapter explicitly identifies gradient descent as a recursively defined sequence of weights.

Connection 2 — Learning Curves

Suppose training loss is:

$$L_1, L_2, L_3, \ldots$$

This is a sequence.

You may observe:

1.2

0.8

0.55

0.40

0.32

0.30

0.299

...

You can ask whether:

$$L_n \to L^*$$

Connection 3 — Numerical Integration

The chapter explicitly describes integrals as limits of convergent sequences.

Conceptually:

Approximation 1

↓

Approximation 2

↓

Approximation 3

↓

...

↓

Integral

Connection 4 — Probability

The chapter connects convergence with the law of large numbers and central limit theorem.

For sample averages:

$$\frac{1}{n}\sum_{k=1}^{n} X_k$$

the law of large numbers describes convergence toward the expected value under its assumptions.

Connection 5 — Algorithm Complexity

Runtime:

$$T(n)$$

can be viewed as a sequence as $n$grows.

Then asymptotic notation describes its long-term growth.

$$\log n < n < n \log n < n^2 < 2^n < n!$$

The chapter directly connects growth comparisons to computer-science time complexity.

---

50. Implementation Cheat Sheet

Sequence

```
def sequence(n):
    return 1 / n
```

Partial sum

```
def series_sum(term, N):
    total = 0.0

    for n in range(1, N + 1):
        total += term(n)
```

```
    return total
```

Geometric series

```
def geometric(q, N):
    total = 0.0
    term = 1.0

    for _ in range(N):
        total += term
        term *= q

    return total
```

Harmonic series

```
def harmonic(N):
    return sum(1 / n for n in range(1, N + 1))
```

Alternating harmonic

```
def alternating_harmonic(N):
    total = 0.0

    for n in range(1, N + 1):
        total += (-1) ** (n + 1) / n

    return total
```

As Nincreases:

$$S_N \to \ln 2$$

---

51. Interview Questions You MUST Know

Q1. What is a sequence?

A sequence is an ordered enumeration of mathematical objects, commonly represented as $\{a_n\}_{n=1}^{\infty}$.

Q2. What does it mean for a sequence to converge?

A sequence converges to $a$if, for every $\epsilon > 0$, all sufficiently large terms lie within $\epsilon$of a.

Q3. Give an example.

$$a_n = \frac{1}{n}$$

and:

$$\lim_{n\to\infty} \frac{1}{n} = 0$$

Q4. What is the difference between sequence and series?

A sequence is a list of terms. A series is the sum of those terms, defined through the limit of its partial sums.

Q5. If $a_n \to 0$, does $\sum a_n$converge?

No.

Counterexample:

$$a_n = \frac{1}{n}$$

because:

$$\frac{1}{n} \to 0$$

but:

$$\sum \frac{1}{n}$$

diverges.

Q6. When does a geometric series converge?

For:

$$\sum q^n$$

it converges when:

$$\boxed{|\ q\ | < 1}$$

and its sum is:

$$\boxed{\frac{1}{1-q}}$$

Q7. What is the difference between Big-O and small-o?

$$b_n = O(a_n)$$

means $b_n$is eventually bounded by a constant multiple of a_n.

$$b_n = o(a_n)$$

means $b_n$becomes negligible relative to a_n.

Q8. Is $\boldsymbol{n = O(n)}$?

Yes.

Is $\boldsymbol{n = o(n)}$?

No.

Because:

$$\frac{n}{n} = 1 \not\to 0$$

Q9. Is $\log n = o(n)$?

Yes.

Because:

$$\frac{\log n}{n} \to 0$$

Q10. Why is the harmonic series important?

Because it demonstrates:

$$a_n \to 0$$

doesn't imply series convergence.

$$\boxed{\frac{1}{n} \to 0 \text{ but } \sum \frac{1}{n} = \infty}$$

Q11. What is absolute convergence?

If:

$$\sum |a_n|$$

converges, then:

$$\sum a_n$$

is absolutely convergent.

Q12. What is conditional convergence?

$$\sum a_n$$

converges but:

$$\sum |a_n|$$

diverges.

Q13. Why does absolute convergence matter?

Because absolutely convergent series can be rearranged without changing their sum.

Q14. What is the squeeze theorem?

If:

$$a_n \leq b_n \leq c_n$$

and:

$$a_n, c_n \to L$$

then:

$$b_n \to L$$

Q15. Why is convergence important in ML?

Because limits underpin differentiation, integration, probability, numerical approximation, and optimization; the chapter specifically connects convergence to gradient descent and stochastic gradient descent.

---

52. 80/20 Priority Map

For your **next AI/ML interview**, I would organize this chapter like this:

Priority	Topic	What you should be able to do
🔴 P0	Sequence	Define + recognize examples
🔴 P0	Limit	Explain intuitively

Priority	Topic	What you should be able to do
🔴	P0 Convergence	Explain ε idea
🔴	P0 Divergence	Recognize
🔴	P0 Series vs sequence	Explain clearly
🔴	P0 Necessary condition $a_n \to 0$	Apply
🔴	P0 Geometric series	Solve
🔴	P0 Harmonic series	Explain divergence
🔴	P0 Big-O	Apply to algorithms
🔴	P0 small-o	Distinguish from Big-O
🔴	P0 Growth rates	Compare functions
🔴	P0 ML convergence	Explain gradient descent connection
🟠	P1 Squeeze theorem	Apply
🟠	P1 Convergence properties	Use
🟠	P1 Comparison test	Apply
🟠	P1 Alternating series	Recognize
🟠	P1 Absolute vs conditional	Explain
🟠	P1 Root test	Recognize/apply
🟡	P2 Countability	Understand
🟡	P2 Cantor diagonal argument	Explain intuition
🟡	P2 $e$representations	Understand
🟡	P2 Basel problem	Know result
🟡	P2 Permutations	Understand
🟡	P2 Cauchy product	Know concept

Priority Topic	What you should be able to do
🟡 P2 Mertens theorem	Know statement

---

53. The 15 Formulas to Memorize

If you have very limited preparation time, memorize these:

1. Number hierarchy

$$\boxed{\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}}$$

2. Sequence convergence

$$\boxed{\lim_{n\to\infty} a_n = L}$$

3. Formal convergence

$$\boxed{|\, a_n - L \,| < \epsilon}$$

eventually.

4.

$$\boxed{\lim_{n\to\infty} \frac{1}{n} = 0}$$

5.

$$\boxed{\lim_{n\to\infty} x^n = 0, \;\; |\, x \,| < 1}$$

6.

$$\boxed{\lim_{n\to\infty} n^{1/n} = 1}$$

7.

$$\boxed{\log n = o(n)}$$

8.

$$\boxed{\log n \ll n \ll 2^n \ll n!}$$

9.

$$\boxed{\sum_{n=1}^{\infty} a_n = \lim_{N\to\infty} \sum_{n=1}^{N} a_n}$$

10. Geometric

$$\boxed{\sum_{n=0}^{\infty} q^n = \frac{1}{1-q}, \ |\, q \,| < 1}$$

11. Harmonic

$$\boxed{\sum_{n=1}^{\infty} \frac{1}{n} = \infty}$$

12. Necessary condition

$$\boxed{\sum a_n \text{ converges} \Rightarrow a_n \to 0}$$

13. p-series

$$\boxed{\sum \frac{1}{n^p} \text{ converges iff } p > 1}$$

14. Alternating harmonic

$$\boxed{\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n} = \ln 2}$$

15. Big-O

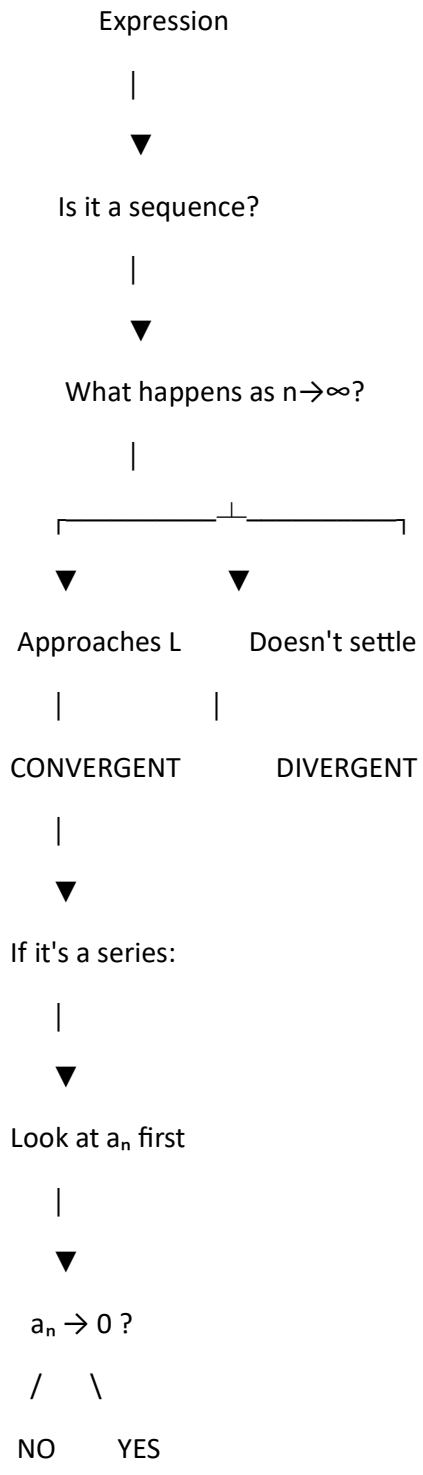
$$\boxed{b_n = O(a_n) \iff |\, b_n \,| \leq C \,|\, a_n \,| \text{ eventually}}$$

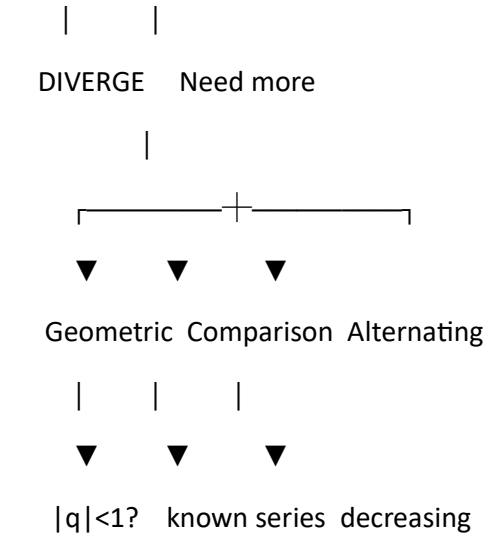
---

54. Final Mental Model

If the interviewer gives you an unfamiliar expression, don't immediately try to memorize a theorem.

Use this reasoning:





And for engineering:

Mathematical sequence

↓

Long-term behavior

↓

Convergence / growth

↓

Algorithm complexity

↓

Numerical approximation

↓

Optimization

↓

Machine Learning

That is the **80/20 core** of this chapter.

The chapter itself concludes that sequences may seem far from everyday ML practice, but they provide foundational structure for quantitative mathematics.